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TimeSkip

*A Game-Based System for Collecting Behavioural Signals and
Estimating ADHD Likelihood*

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February 2026



Acknowledgements

We would like to express our sincere gratitude to our supervisor, **Prof. Erel Segal-Halevi**, for his professional guidance, continuous support, and constructive feedback throughout the project.

We also thank the **Department of Occupational Therapy at Ariel University** and the collaborating students—Michal Leff-Cohen, Hila Buchbut, and Shira Shalit—for their valuable clinical perspective and their contribution to shaping the game’s objectives, mechanics, and child-appropriate design decisions.

We gratefully acknowledge the support of **Rambam Elementary School in Haifa**, and especially Shlomit Guzman, for enabling and assisting the playtesting process in a real educational setting. Finally, we thank the participating children for their time and cooperation during the testing sessions; their gameplay and feedback were essential for improving usability and overall player experience.

Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) is commonly associated with persistent difficulties in sustained attention, inhibitory control, and working-memory performance. Alongside clinical assessment and intervention, there is a growing need for accessible, engaging tools capable of capturing objective behavioural indicators in naturalistic settings.

This project presents **TimeSkip**, a short, story-driven game designed to unobtrusively collect behavioural signals during gameplay and use them to estimate ADHD likelihood via a machine-learning classifier. The player progresses through a guided sequence of scenes: a brief controls tutorial, a forest task focused on goal-directed collection under hazard constraints, a working-memory task inside the dwarf’s house, and a river-crossing task that requires timing and precision.

Throughout play, the system records fine-grained in-game events and aggregates them into a fixed numeric feature vector that summarizes performance outcomes, reaction-time statistics, and input dynamics. These features are then provided to a trained classification model, which returns an inference result intended to support research and early screening workflows rather than replace formal diagnostic procedures.

This report details the game design and narrative rationale, the data logging and feature extraction pipeline, the technical system architecture, and the evaluation approach conducted in a school-based playtesting setting.

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Chapter 1

Introduction

1.1 Motivation

We live in a hyper-digital era where children are accustomed to high-paced, interactive, and visually stimulating environments. Recent studies indicate that prolonged exposure to such media correlates with changes in cognitive processing styles, creating a generation of "Digital Natives" who engage differently with the world around them [1].

However, the world of ADHD assessment has not kept pace with this cultural shift. The motivation for this project arose from a simple observation: existing diagnostic tools are often static, monotonous, and disconnected from the child's daily reality. A child asked to perform a repetitive task for 20 minutes may fail not because they cannot focus, but because they have no intrinsic motivation to do so. This "Engagement Gap" creates a situation where the tool measures the child's tolerance for boredom rather than their true attention capacity.

"TimeSkip" was born out of a unique academic collaboration at Ariel University between the Department of Computer Science and the Department of Occupational Therapy. Our shared vision was to bridge this gap by creating a tool that speaks the child's language. By wrapping clinical assessment mechanics inside a narrative-driven video game, we aim to transform the diagnostic process from a stressful chore into an engaging experience, allowing us to capture the child's natural cognitive performance.

1.2 Problem Statement

While digital culture has evolved, the core methodologies for ADHD screening have remained largely static, suffering from three fundamental scientific limitations:

1. **Low Ecological Validity:** Continuous Performance Tests (CPTs) measure attention in a sterile, laboratory-like environment. Research indicates that performance in these isolated settings often correlates poorly with a child's functioning in real-world, multisensory environments (e.g., a noisy classroom) [2]. The data obtained is thus "clean" but often clinically irrelevant to the child's daily challenges.
2. **Observer Bias and Inter-Rater Reliability:** The most common diagnostic tool—behavioral checklists—depends entirely on subjective reporting. Studies consistently show signifi-

cant discrepancies between parent and teacher reports, creating a "noisy" clinical picture [3]. This subjectivity makes it difficult to establish an objective baseline for the child's condition.

3. **The "White Coat" Effect:** The formal testing environment itself can act as a confounding variable. The stress of being tested in a clinic can induce "test anxiety," masking typical behavioral patterns or causing performance drops unrelated to attention deficits.

1.3 The Solution: "TimeSkip"

"TimeSkip" addresses these methodological gaps by implementing a *Stealth Assessment* approach. Instead of a formal test, the system offers a dual-layer solution:

- **The Narrative Layer (Frontend):** A gamified environment designed in collaboration with Occupational Therapy. By masking the assessment as a quest to repair a time machine, the system bypasses the "White Coat" effect, allowing the child to behave naturally and reducing test anxiety.
- **The Data Layer (Backend):** While the user plays, the system unobtrusively captures fine-grained biomarkers such as Inhibition (holding back a response in the River level) and Working Memory (recalling sequences in the Dwarf level).

Clinical Goal: It is crucial to clarify that TimeSkip is not a replacement for professional diagnosis. Rather, it acts as an objective *Preliminary Screening Tool*. Its output is a probability score designed to flag at-risk children and support the decision-making process for referral to comprehensive clinical evaluation.

1.4 Project Objectives

The overarching goal of this project was to translate abstract clinical concepts into concrete, playable game mechanics. Specifically, the project aimed to design and implement a continuous game loop comprising three distinct assessment modules:

1. **Objective 1: Visual Discrimination & Impulsivity (The Forest Scene).** To design a dynamic environment where the player moves horizontally and encounters a "Giant Rock" portal leading to a high-speed challenge. The objective was to create a mechanic requiring the user to collect target items (Time Machine Screws) while avoiding distractors that visually resemble the targets (Poisonous Berries). This module aims to measure Inhibition by analyzing whether the child "panic clicks," jumps excessively, or fails to verify the object type before collecting.

2. **Objective 2: Working Memory Assessment (The Dwarf's House).** To implement a memory-based interaction within the narrative. Upon reaching the "Dwarf's House," the player engages in a sequence recall task based on images and situations. The objective here is to isolate and measure the child's Working Memory capacity distinct from their motor reflexes.
3. **Objective 3: Motor Planning & Stress Regulation (The River Crossing).** To develop a complex navigation challenge where the player must cross a river by hopping between rocks while the screen scrolls forward (forcing movement). The objective includes introducing dynamic threats (Predatory Fish jumping from the water) to test the child's ability to balance speed with caution—requiring calculated logic rather than impulsive rushing.
4. **Objective 4: Narrative Synthesis.** To bind these mini-games into a cohesive story where success leads to the (theoretical) repair of the Time Machine, providing the child with a sense of purpose and closure.

Chapter 2

Related Work and Theoretical Background

2.1 The Neurocognitive Basis of ADHD

Attention-Deficit/Hyperactivity Disorder (ADHD) is fundamentally a disorder of Executive Functions (EF) – the top-down mental processes that enable self-regulation and goal-directed behavior. While the diagnostic criteria (DSM-5) focus on external symptoms like fidgeting or inattention, the underlying neurological mechanism is best explained by Barkley's Unifying Theory of ADHD [4].

According to this model, the core deficit in ADHD is not "attention" per se, but rather a failure in Response Inhibition. This primary deficit disrupts four secondary executive functions, with Working Memory being the most critical for academic performance.

2.1.1 Response Inhibition (The "Brake" System)

Response inhibition refers to the ability to suppress a dominant, automatic response in favor of a more appropriate, goal-directed action. It involves three interrelated processes:

- Inhibiting the initial "pre-potent" response (stopping oneself from acting impulsively).
- Stopping an ongoing response (ceasing an action when feedback indicates it is wrong).
- Interference control (protecting a period of delay from distraction).

Scientific implication: An effective screening tool must test the subject's ability to "hold back" a reaction under time pressure (a Go/No-Go paradigm) rather than just measuring their ability to react quickly.

2.1.2 Working Memory (The "Mental Workspace")

Based on Baddeley's model [5], Working Memory is the system responsible for temporarily holding and manipulating information. It is divided into the Phonological Loop (auditory information) and the Visuo-Spatial Sketchpad (visual information). Children with ADHD often show significant deficits in the ability to retain sequences or follow multi-step instructions, as their "mental workspace" is easily overwritten by external distractions. **Scientific implication:** To accurately detect ADHD, a system must measure the capacity of this workspace (e.g., how many items can be recalled in sequence) independently of motor skills.

2.2 Game-Based Assessment: Stealth Assessment vs. Gamification

While the use of games in healthcare is growing, it is essential to distinguish between standard "Gamification" and the "Stealth Assessment" methodology employed in this project.

2.2.1 The Concept of Stealth Assessment

Coined by Valerie Shute [6], Stealth Assessment refers to the process of weaving psychometric testing directly into the fabric of a game. Unlike traditional tests where the user stops to answer questions (disrupting flow), Stealth Assessment analyzes the user's natural interactions—such as mouse movement velocity, hesitation before a jump, or error correction patterns—to infer competency in real-time.

In "TimeSkip," this is achieved by mapping specific game mechanics to cognitive constructs:

- **Gamification approach (What we avoided):** Adding points or badges to a standard boring test (e.g., "Click the button 100 times to get a star").
- **Stealth Assessment approach (What we built):** The child believes they are crossing a river to escape a fish, but the system is actually measuring their Inhibitory Control by logging the milliseconds of delay before a jump.

2.2.2 The "Fun-First" Paradigm

A critical theoretical advantage of game-based assessment is the reduction of Test Anxiety. Clinical literature suggests that when a child knows they are being evaluated, their performance may degrade due to stress (the "White Coat Effect") [7]. By masking the assessment as a narrative quest, the system captures a more authentic representation of the child's cognitive abilities in a low-stress environment.

2.3 The Gap: Ecological Validity and Dynamic Complexity

While Chapter 1 discussed the user experience issues (boredom), this chapter addresses the scientific flaw in current tools: the lack of Ecological Validity. Standard Continuous Performance Tests (CPTs) isolate a single variable in a sterile environment. The child sits in a quiet room and reacts to a blinking light. However, real-world attention deficits do not manifest in silence; they appear when a child tries to pack their bag while the TV is on, or listen to a teacher while a classmate is talking.

Barkley's Theory of ADHD argues that the core deficit is not just "staring at a screen," but Goal-Directed Persistence in the face of distractions [4]. Therefore, a valid assessment tool

must mimic this complexity. It cannot be static; it must be dynamic. It must force the user to prioritize targets (screws) over distractors (berries) while managing movement—simulating the cognitive load of the real world.

2.4 Comparative Analysis: Static vs. Dynamic Assessment

To understand where "TimeSkip" fits in the clinical landscape, we must compare the nature of the data collected, not just the user experience.

Feature	Standard CPTs (e.g., MOXO/TOVA)	TimeSkip (Stealth Assessment)
Stimulus Type	Abstract, 2D Geometric shapes (Static).	Narrative, physics-based environment (Dynamic).
Cognitive Load	Low: Single-task focus (press space).	High: Multi-tasking (navigate + collect + avoid).
Inhibition Measure	Binary: Did they click the wrong button?	Continuous: Did they hesitate? Did they change direction mid-jump?
Subject State	Conscious of being tested ("Performance Mode").	Immersed in flow ("Natural Mode").

Table 2.1: Comparative Analysis: Standard CPTs vs. TimeSkip

Conclusion of Theoretical Background: The "Holy Grail" of ADHD assessment is to observe the child’s brain at work without the interference of stress or artificiality. TimeSkip does not aim to merely "gamify" a test; it aims to create a transparent window into the child’s cognitive functioning. By removing the walls of the "test," we allow the child’s authentic behavioral signature—their true impulsivity and true memory capacity—to emerge naturally and be measured with digital precision.

Chapter 3

Game Design and Narrative

3.1 Game Story and Concept

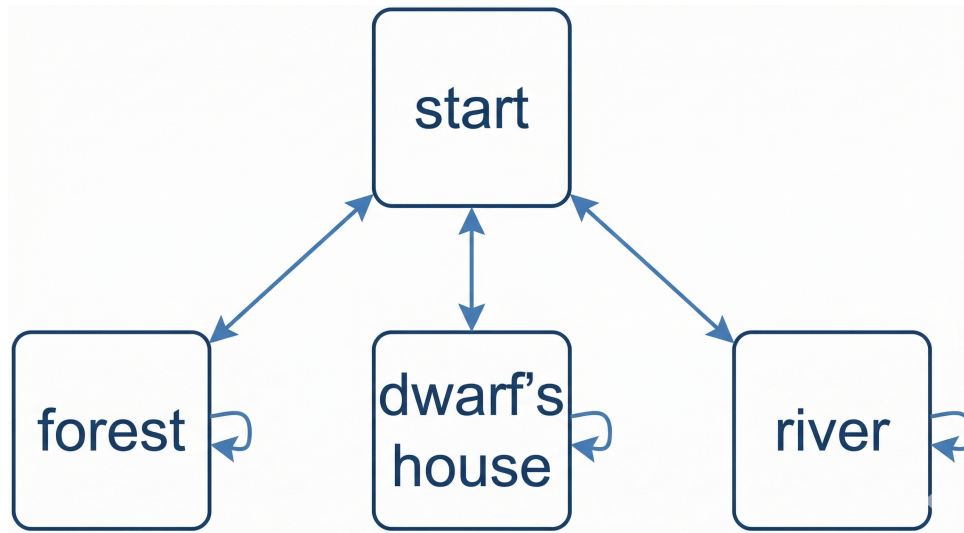
TimeSkip is framed as a short, guided adventure that quietly measures behavior while the player tries to "help fix a time machine." The narrative glue is simple and purposeful: the player collects missing screws in the forest, completes a memory challenge inside the dwarf's house, and then crosses a dangerous river to continue forward. This keeps the session feeling like one connected journey instead of unrelated tasks. The intended flow was also shaped with occupational-therapy input.

3.2 User Journey and Flow

The game is designed as a linear, single-player session:

1. **Start scene (controls tutorial)** – quick warm-up to confirm the player can jump and move.
2. **Forest** – collect screws while avoiding hazards.
3. **Dwarf's House (Memory Game)** – short working-memory trials.
4. **River** – cross by jumping between rocks, avoiding fish and staying in bounds.

Even though the progression is linear, the player can return to the Start scene from the end-of-run UI (useful after failing and retrying). Both the forest and river outcome panels include a "Start" button.



3.3 Minigame 1 – Start Scene (Controls Tutorial)

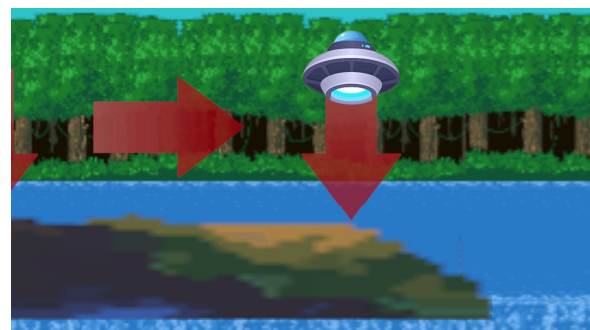
The Start scene is a controls tutorial (not a scored minigame). Its only goal is to ensure the player understands the basic controls before entering gameplay areas.

- First prompt: "Use space key to jump."
- After one Space press, the prompt changes to: "Use right arrow to go right or left arrow to go left."
- After pressing Left or Right once, the tutorial completes and the prompts disappear.

There is no win/lose, no timer, and it does not contribute to the 23-feature prediction vector. Although this scene does not contribute to the prediction model, it serves a critical role in 'Base-lining' the user's motor capability, ensuring that subsequent errors are cognitive, not mechanical.



(a) Jump Prompt



(b) The Spaceship (End Goal)

Start scene prompt screenshots.

3.4 Minigame 2 – Forest (Collecting Screws)

The forest section is framed as: collect screws to fix the time machine while avoiding hazards.

- **Start of run:** when the player enters the area, an instruction appears once per session: "Collect the screws to fix the time machine! Avoid the berries, they are poisonous! Press any key to start."
- **Controls:** Left/Right arrows to move, Space to jump.
- **Objectives and hazards:**
 - Win condition: collect all screws (when collected = configured total).
 - Lose condition: collide with a poisonous berry, or go out of camera bounds.
- **What is logged / measured:**
 - Item lifecycle: `item_spawn` and `item_resolve` for screws/berries.
 - Input dynamics: keypress count and inter-key gap mean/CV.
 - Reaction time: median + CV for target actions (e.g., pickups).



Forest gameplay screenshot showing screws, progress UI, and berry hazards.

3.5 Minigame 3 – Dwarf's House (Memory Game)

The dwarf's house is a working-memory task: the player sees a snapshot of a scene, then answers a multiple-choice question about it; this contributes reaction-time and accuracy metrics. Theoretically, this task isolates the 'Visuo-Spatial Sketchpad' component of Working Memory, a core executive function outlined in Baddeley's model [8].

Flow (core loop):

1. Show snapshot for 5 seconds.
2. Hide snapshot, show question + answer options.
3. The player selects an answer (mouse click).
4. Correct → next level; wrong → lose a life. The player starts with 3 lives.

Completion / end conditions

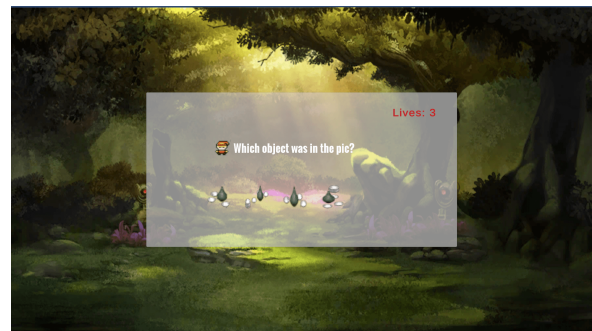
- Successful completion can occur by reaching a configured level milestone (e.g., level index 4).
- Failure occurs if lives drop to 0 → game over panel.

What is logged / measured

- Lifecycle: `dwarf_minigame_start`, `dwarf_minigame_end` with end reason.
- Per-level / per-trial events: `memory_level_start`, `dwarf_trial_start`, `dwarf_trial_response` (includes correctness + reaction time in ms from question onset), and `memory_level_result`.
- Aggregated features used later: `dwarf_enters`, `dwarf_wm_rt_median`, `dwarf_wm_rt_cv`, `dwarf_correct`, `dwarf_incorrect`.



(a) Snapshot Phase



(b) Question Phase

Dwarf's House UI screenshots showing the memory task flow.

3.6 Minigame 4 – River (Jumping Across Rocks)

The river section is a navigation challenge: **jump between rocks to reach the far bank**, while avoiding hazards and staying within camera bounds.

Entry + start sequence

- The player approaches the portal and sees: “Click enter to go to the river.” Press **Enter** to load the river scene.

- The river run begins only after instructions complete; then a **countdown** appears (default **5 seconds**). Default run duration is **60 seconds** if not ended earlier by win/lose.

Controls

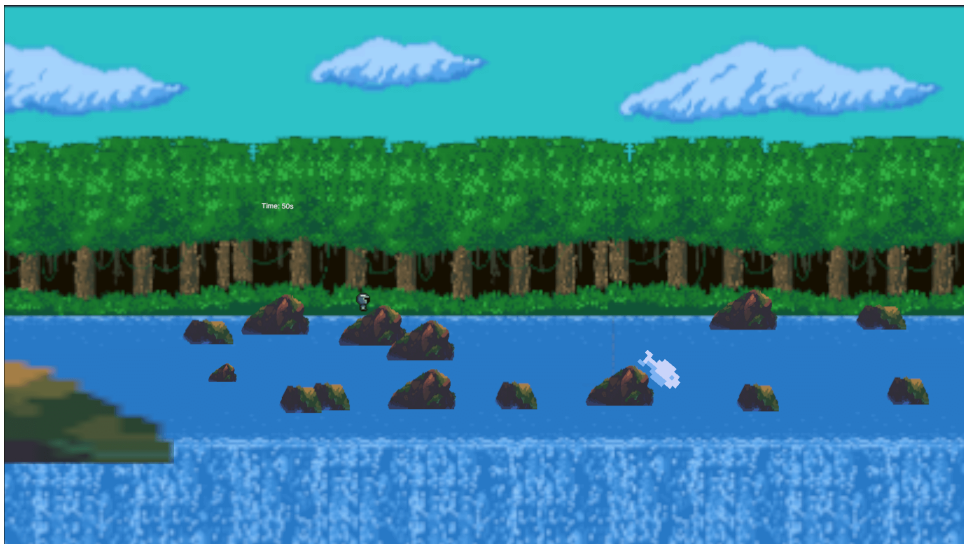
- Left/Right arrows: move on rocks
- Space: jump (logs jump start/land)
- Down arrow: fall faster (temporary higher gravity)

Win / lose

- **Win:** collide with the far bank end object → logs `river_finish` “win” ends run.
- **Lose:** collide with fish, or go out of camera bounds ends run.

What is logged / measured

- Jump events: `jump_start` and `jump_land` tied by a generated jump id.
- Hazards/outcomes: fish collisions and `river_finish` result; out-of-bounds losses also logged.
- Approaches toward fish (proximity-based).
- Input dynamics: keypress count and inter-key gap mean/CV while in the river.



River gameplay screenshot showing rocks and fish hazards.

3.7 Design Considerations

A core goal in the game design was to keep the experience engaging enough to feel like a real adventure, while still creating situations that reveal meaningful differences in attention, impulse control, and working-memory performance.

Maintaining attention and engagement

To reduce boredom and keep players focused, each section has a clear short-term goal and immediate feedback. In the Forest, the screw-collection objective is always visible through the progress UI and the run ends as soon as the goal is achieved (instead of continuing unnecessarily). In the River, the countdown and time-limited run create urgency and keep the pace active rather than repetitive. The overall flow also benefits from being broken into distinct environments (Forest → Dwarf's House → River), so the player's attention is "refreshed" by a change in setting and task type.

Screws and berries intentionally look similar

In the Forest, screws and berries were designed to be visually close enough that the player cannot rely on "lazy recognition." This creates a controlled attention challenge: the player must stay alert and actually inspect what they are approaching, instead of mindlessly grabbing everything. The similarity pushes selective attention and makes impulsive movement more costly—especially under time pressure or excitement—without requiring complex rules.

This mechanic mirrors the classic 'Go/No-Go' paradigm used in clinical settings to measure response inhibition, requiring the subject to distinguish between targets and non-targets under time pressure [9].

Impulse control through hazards and risk

Impulse control is supported mainly through consequences and temptation zones. Berries act as a simple hazard that encourages the player to slow down and decide before jumping or moving forward. Out-of-bounds failure also discourages uncontrolled movement. In the River, fish hazards and the risk of falling out of bounds produce similar "think-before-you-act" moments, while still keeping the controls simple (left/right + jump). The design tries to create situations where rushing is tempting—but not always optimal.

According to Barkley's unifying theory of ADHD, this inability to pause and assess risk ('Action Restraint') is the primary behavioral deficit in hyperactive-impulsive profiles [4].

Working memory depends on visibility, not trickiness

In the Dwarf's House memory task, the goal is to measure working-memory ability—not visual guessing. Because of that, the important details in the snapshot have to be very large and easy to perceive. If the details are small, the task becomes about eyesight, screen quality, or luck, and it stops reflecting memory performance. Making key items visually big reduces noise

and ensures that forgetting is the main difficulty, not failing to notice.

Difficulty tuning: fun, not boring and not too hard

Forest and River difficulty was tuned around the idea of “fast enough to be exciting, slow enough to be manageable.”

If movement is too slow or hazards too rare, players drift into autopilot and the session becomes boring.

If movement is too fast or hazards too punishing, the player fails repeatedly and frustration replaces meaningful behavior.

So the pace was adjusted until most players can succeed with focus, while still making mistakes possible when attention drops or impulses take over. The River also uses a countdown before the run begins, which helps players prepare and reduces “instant fail” confusion.

Clear instructions: written + vocal

Because the target audience is children, instructions are delivered in two ways: on-screen text and voice prompts. The Start scene teaches controls step-by-step (jump first, then movement), the River, and the Forest uses a one-time instruction prompt before beginning. This reduces reliance on reading ability alone and lowers the chance that misunderstanding rules will dominate the gameplay data.

Chapter 4

Data Logging and Processing

A central goal of TimeSkip is to collect objective, fine-grained behavioral signals during a short play session and convert them into a compact representation that can be stored and sent to a prediction service. During gameplay, the system records structured events; when the session ends, these events are aggregated into (1) a single human-readable “summary line” and (2) a fixed 23-value numeric feature vector for the classifier.

4.1 Event Logging Framework

Logging approach. The game uses an event-based logger that records structured events while the player interacts with each scene. Export occurs when a session concludes, triggered by the player’s arrival to the spaceship.

Events are organized by area/scene and cover both outcomes (win/lose conditions) and “process” signals such as movement patterns, proximity behavior, and input dynamics.

Event categories captured. Across the different areas, the logger records:

- **Item lifecycle (Forest):** `item_spawn` and `item_resolve` for screws/blueberries.
- **Collisions and outcomes (Forest/River):** hazard collisions, out-of-bounds, and end-of-run results.
- **Approach behavior (Forest/River):** proximity-based “approach” events toward hazards (e.g., fish or blueberry).
- **Input dynamics (Forest/River):** number of key presses and inter-keypress gap statistics (mean and coefficient of variation). High-frequency keylogging allows the system to capture ‘micro-movements’ and fidgeting behavior, providing an objective proxy for the hyperactivity symptom cluster defined in the DSM-5 [11].
- **Jump events (River):** `jump_start` and `jump_land`, linked via a generated jump id.
- **Memory game lifecycle and trials (Dwarf’s House):** minigame start/end, per-level start, trial presentation, and trial response events including correctness and reaction time.

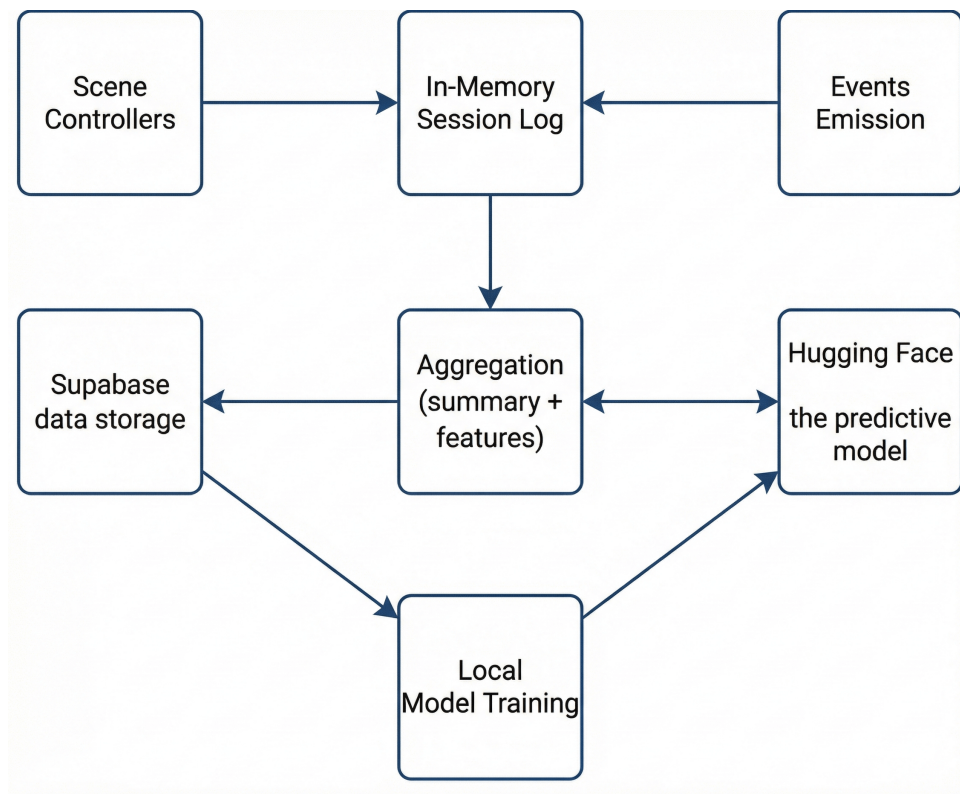


Figure 4.1 — Logger architecture.

4.2 Session Summary and Feature Extraction

From events to a compact representation. At the end of the session, the logger produces:

1. a one-line summary for storage, and
2. a 23-element numeric feature vector for model inference.

In the current build, the Supabase record is intentionally minimal: by request, only the summary line is included in the Supabase row body.

Important consistency rule (shared ordering). The summary line and the 23-value feature vector contain the same information in the same order—the only difference is formatting (human-readable line vs. numeric array).

How key statistics are computed (high level).

- **Counts** (e.g., collisions, out-of-bounds, key presses) are summed over the session.
- **Reaction-time medians** are computed by collecting per-event latencies and taking the median.
- **Coefficient of variation (CV)** is computed as (standard deviation / mean) for the relevant timing series (reaction times or inter-keypress gaps). This specific metric was prioritized

because clinical literature identifies ‘Intra-Individual Variability’ (IIV)—inconsistency in response times—as one of the most robust cognitive markers for ADHD, often yielding higher diagnostic utility than mean reaction time alone [10].

- **Inter-keypress gap mean/CV** are computed from the time differences between consecutive key presses within the relevant area.

Table 4.1: Feature vector definition (23 values)

Index	Feature Name	Description (what it captures)
0	<code>screws_spawned</code>	Number of screw spawn events during Forest.
1	<code>screws_collected</code>	Number of screws successfully collected in Forest.
2	<code>forest_fish_collisions</code>	Hazard collisions in Forest (in practice, this corresponds to berry collisions in the Forest implementation).
3	<code>forest_out_of_bounds</code>	Number of Forest losses due to leaving camera bounds.
4	<code>forest_5_screws_collected</code>	Indicator/count tied to the “5 screws collected” milestone in Forest.
5	<code>forest_approaches</code>	Number of proximity “approach” events toward hazards in Forest.
6	<code>target_rt_median</code>	Median reaction time for target actions (e.g., pickups) derived from the session log.
7	<code>target_rt_cv</code>	Coefficient of variation of target reaction times.
8	<code>forest_keys</code>	Total key-press count while in Forest.
9	<code>forest_key_gap_mean</code>	Mean time gap between consecutive key presses in Forest.
10	<code>forest_key_gap_cv</code>	CV of inter-keypress gaps in Forest.
11	<code>river_fish_collisions</code>	Number of fish collision events in River.
12	<code>river_out_of_bounds</code>	Number of River losses due to leaving camera bounds.
13	<code>river_win</code>	Win indicator for River (reach the far bank).
14	<code>river_approaches</code>	Number of proximity “approach” events toward fish in River.

Index	Feature Name	Description (what it captures)
15	<code>river_keys</code>	Total key-press count while in River.
16	<code>river_key_gap_mean</code>	Mean time gap between consecutive key presses in River.
17	<code>river_key_gap_cv</code>	CV of inter-keypress gaps in River.
18	<code>dwarf_enters</code>	Number of Dwarf's House entries/rounds contributing to WM metrics.
19	<code>dwarf_wm_rt_median</code>	Median working-memory response time (ms) measured from question onset.
20	<code>dwarf_wm_rt_cv</code>	CV of working-memory response times.
21	<code>dwarf_correct</code>	Number of correct answers in Dwarf's House trials.
22	<code>dwarf_incorrect</code>	Number of incorrect answers in Dwarf's House trials.

(Note: The Start Scene tutorial is onboarding only and does not contribute to the 23-value vector.)

4.3 Data Storage (Supabase)

What is stored and why. At session end, a compact export is uploaded to Supabase via a REST call. The design prioritizes privacy: no raw screenshots or personal data are uploaded by default, and the prediction API receives only numeric features.

Stored fields: `session_id`, `session_json`

Current build note: only the minimal summary line is included in the Supabase row body.

id	created_at	player_name	session_id	session_json
121	2026-01-28 09:40:18.767282+0	EMPTY	ca2f5ea8-fa4d-444b-a0e5-9f58dd95a511	{"summaryLine":{"screws_spawned=156, s
122	2026-01-28 09:41:34.647348+0	EMPTY	94040eb4-3720-458d-b431-a3873f993bc	{"summaryLine":{"screws_spawned=135, s
123	2026-01-28 09:58:52.411111+00	EMPTY	703cc7d4-732c-4a4d-ba80-e2a38d7fb36f	{"summaryLine":{"screws_spawned=0, scr
124	2026-01-28 10:03:55.656952+0	EMPTY	8e976380-22e9-4155-bd02-56fe0d059772	{"summaryLine":{"screws_spawned=65, sc
125	2026-01-28 10:08:34.747928+0	EMPTY	b259abbc-727f-439e-971b-0eb1d6e3183a	{"summaryLine":{"screws_spawned=76, sc
126	2026-01-28 10:11:30.528586+00	EMPTY	b77df758-b59e-439f-ab40-851f03b75532	{"summaryLine":{"screws_spawned=0, scr
127	2026-01-28 10:28:58.914545+01	EMPTY	679bd344-d09e-4965-ab7b-fb56fa3c32b	{"summaryLine":{"screws_spawned=42, sc
128	2026-01-28 10:31:38.486506+00	EMPTY	ebbf4851-b4dc-46f9-9fef-2b4e8d1561d4	{"summaryLine":{"screws_spawned=70, sc
129	2026-01-28 10:32:13.843072+00	EMPTY	eb1551e3-d313-485d-997e-91ee29fb3648	{"summaryLine":{"screws_spawned=68, sc
130	2026-01-28 10:41:00.585572+01	EMPTY	1a3490a9-8563-42c6-8c94-5b122a181e1d	{"summaryLine":{"screws_spawned=148, s
131	2026-01-31 13:42:08.607289+01	EMPTY	63f7cf61-bfb8-4b08-90a5-35c68b711550	"screws_spawned=15, screws_collected=4, forest_fish_collisions=0, forest_out_of_bounds=2, forest_5_screws_collected=0, forest_approaches=12, target_rt_median=6.21, target_rt_cv=0.48, forest_keys=83, forest_key_gap_mean=0.73, forest_key_gap_cv=2.34,
132	2026-01-31 13:46:35.974788+00	EMPTY	fd4465b6-e5d7-4eef-bf9c-a8c69410d5fd	
133	2026-02-01 10:15:34.189962+00	EMPTY	efec82fb-63d8-4dfd-8373-c0184ea5f858	
134	2026-02-01 10:19:25.298022+01	EMPTY	b71ae222-583c-4437-a888-1eefed57d315	
135	2026-02-01 10:36:27.396814+01	EMPTY	45840693-05e6-46a4-ac47-209a93b92d2	
136	2026-02-01 10:38:50.03314+00	EMPTY	26c83102-b00d-486e-88ca-9cf330936ee	
137	2026-02-01 11:42:17.808886+00	EMPTY	1d1a5e53-d4cc-4145-9e5c-57c22aeaf0aa	
138	2026-02-01 13:12:02.858358+01	EMPTY	60bf1b7b-60f8-405b-aaaa-136f3f589901	

Figure 4.3 — Supabase DB.

4.4 Model API Integration

Inference call. The logger also sends the 23-value feature vector to a prediction endpoint (a Hugging Face router).

Request format and security. The payload is a JSON object containing a single row of features under "data". A 20-second timeout is used to avoid blocking the session end flow.

Response handling and on-screen feedback. The game overlays a short two-line prediction message: either “ADHD detected.” or “No ADHD detected.” with the corresponding probability. The overlay is informational and does not change gameplay or exports.

Chapter 5

System Architecture & Predictive Model

5.1 Tools and Technologies

The game was developed in the Unity Engine using a collection of custom C# scripts that control player movement, scene logic, UI prompts, and (most importantly for this project) structured event logging. During play, the game records fine-grained gameplay events and then converts them into two outputs at the end of the session: (1) a compact session export intended for storage and later analysis, and (2) a fixed-length numeric feature vector used for machine-learning prediction.

For data storage, the project uses Supabase as a lightweight backend. At session end, a minimal summary JSON is uploaded to Supabase (in this build, the upload emphasizes a human-readable “summary line” and a compact session structure rather than raw logs). This approach keeps the back end focused on analytics while keeping gameplay fully client-side.

Model training is done in Python using scikit-learn. The training pipeline is saved as a serialized file (`adhd_model.pkl`) and the deployment expects the model’s input dimensionality to remain consistent across versions (this classifier is trained to expect 23 features per session).

For hosting the trained model as an API, the project supports practical deployment style on Hugging Face Spaces - a FastAPI service that exposes `POST /predict` for stable programmatic access.

5.2 System Architecture

At a high level, the system separates front-end gameplay (Unity) from back-end analytics and prediction (Supabase + model API). During gameplay, events are accumulated locally. When the session ends, Unity triggers an export routine that sends:

- a session export to Supabase (for storage and research analysis), and
- a numeric feature vector to the model API (for real-time prediction).

A key design choice is that the online services do not need to know anything about Unity’s runtime logic: the backend receives only (a) compact structured summaries and (b) numeric features. The default configuration explicitly avoids uploading raw screenshots or personal data; the model endpoint receives only numbers.

- **Supabase data upload.** The client uploads via Supabase’s PostgREST endpoint using a standard HTTP POST request with the project’s URL/table configuration and the expected authorization headers. The uploaded row includes identifiers such as `player_name` and `session_id`, and a `session_json` export that begins with a one-line summary and can include scene-keyed event sections (in this build, only the minimal summary line is included by request).
- **Model API request/response.** If a model endpoint is configured, Unity also sends a JSON payload in the form `{"data": [[f0, f1, ..., f22]]}` (optionally including an `fn_index` for some routed endpoints), with an optional bearer token for private deployments. The client applies a bounded request time (20 seconds) and logs a short diagnostic preview of responses. When the response matches the expected compact schema, the game can briefly display a two-line overlay such as “ADHD detected.” / “No ADHD detected.” along with the probability, without changing gameplay behavior.

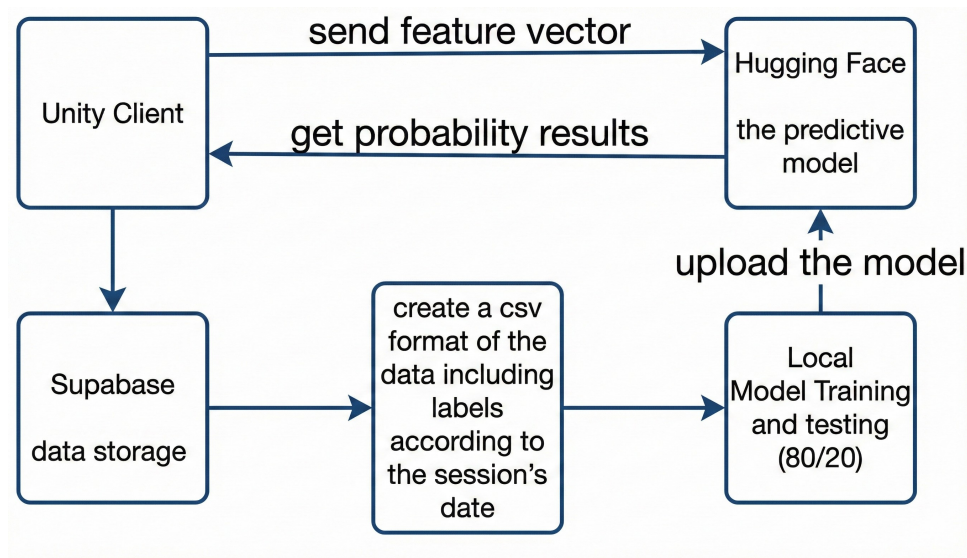


Figure 5.1: Block diagram of the system architecture.

5.3 Implementation Details per Scene

The implementation follows a consistent pattern: each gameplay area has scripts responsible for (1) player control and interactions, (2) UI prompts and run lifecycle, and (3) event logging through a shared experiment logger. The logger and its data models are centralized so that each scene contributes events in a consistent structure.

Start Scene (Controls Tutorial)

The Start scene is implemented as a short onboarding step that presents sequential on-screen prompts (“Use space key to jump”, then “Use right arrow... or left arrow...”). The logic is managed by `InstructionManager.cs`, which marks steps as completed based on key presses.

Forest (Screw Collection and Hazard Interaction)

In the forest area, movement, item tracking, and logging are split across dedicated scripts. The main control and tracking components include:

- `PlayerMovementForest.cs` (movement and control handling),
- `ForestItemTrack.cs` (item lifecycle tracking such as screw/blueberry),
- `CollisionHandler.cs` and `UIManager.cs` (shared win/lose handling and progress UI), and
- `ExperimentLogger.Forest.cs` plus related logger modules for approaches and scene events.

River (Jumping Across Rocks)

The river scene uses a controller-driven run lifecycle with a countdown and a fixed duration, managed by `GameController.cs`. UI state (countdown, timer, outcome panels) is handled by `UIManager.cs`, while collisions and outcomes are handled by `CollisionHandler.cs`. Player motion and jump logging are handled by `PlayerMovementRiver.cs`, and the area's event stream is recorded through `ExperimentLogger.River.cs`.

Dwarf's House (Working-Memory Minigame)

The memory minigame is implemented as a structured level/trial loop (snapshot display → question/options → response → life/level update). Core logic is managed by `GameManager.cs`, UI by `UIManagerMG.cs`, and content/question handling by `QuestionManager.cs`. The export routine (used to persist results and prepare features) is connected through the shared persistence logger.

5.4 Model Development and Training

The classifier was trained using a supervised learning approach with a logistic regression model implemented in scikit-learn. Each gameplay session was converted into a fixed-length numerical feature vector summarising behaviour across scenes (forest, river, dwarf). The target label is binary (ADHD = 1, No-ADHD = 0), and the model learns to estimate the probability that a session belongs to the ADHD class.

Training was performed using a scikit-learn Pipeline to ensure that the exact same preprocessing is applied consistently during both training and evaluation. The pipeline included two sequential stages:

Feature scaling (StandardScaler)

All numeric features were standardised to zero mean and unit variance. This step is required for logistic regression because the model optimises a linear decision boundary and is sensitive

to feature scale; standardisation prevents features with larger numeric ranges (e.g., key counts) from dominating the optimisation. This preprocessing step is a fundamental requirement for gradient-based estimators to ensure convergence and prevent bias towards features with larger raw magnitudes [13].

Logistic regression classifier

Logistic regression was used as the final estimator because it produces a calibrated probability output via the logistic (sigmoid) function, allowing the system to present results as a percentage likelihood of ADHD rather than only a hard classification. Studies comparing machine learning models for ADHD diagnosis have shown that linear models like Logistic Regression often perform on par with more complex models for tabular behavioral data, while offering superior interpretability [12].

To evaluate generalisation performance, the dataset was split into training and test subsets using an 80/20 hold-out split. The model was fit only on the training portion. Predictions were then generated for the test subset, and performance metrics were computed from these held-out predictions. The reported results include overall accuracy and class-specific error types (true positives, true negatives, false positives, false negatives), and the probability outputs were further examined to understand model confidence across correct and incorrect predictions.

To summarise performance on the held-out test set, the model was assessed using the confusion matrix (ADHD vs No ADHD), which reports the number of correct and incorrect classifications for each class. In the current results, the model's errors appear only in one direction: it misclassified some No-ADHD sessions as ADHD (false positives), while it did not misclassify ADHD sessions as No-ADHD (no false negatives). From an application perspective, this is the safer failure mode: in the worst case, the system may recommend formal diagnostic follow-up for a child who does not ultimately meet diagnostic criteria, rather than failing to flag a child who may require further evaluation.

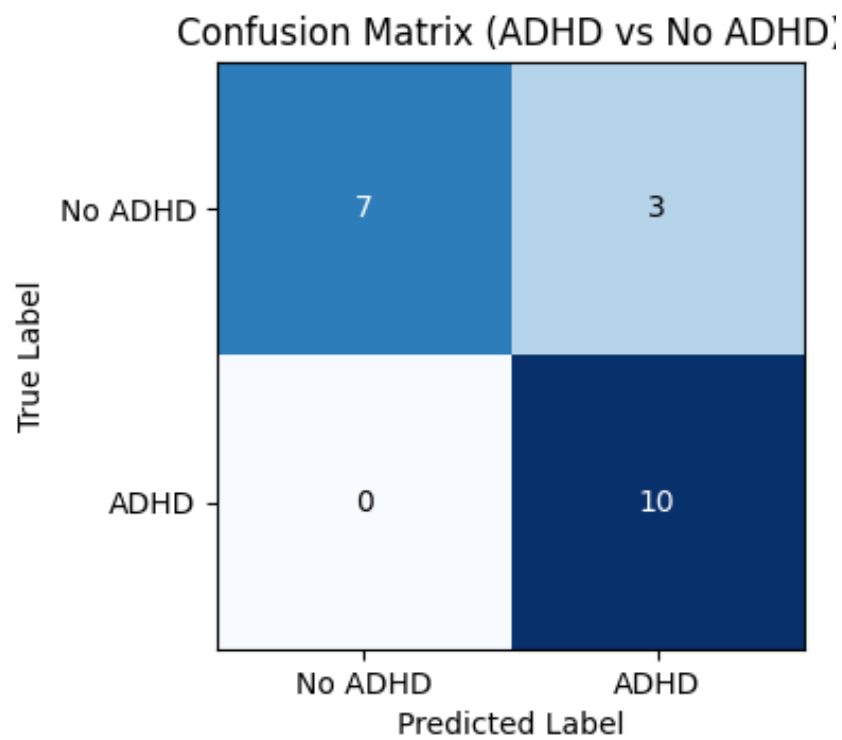


Figure 5.4: Confusion matrix for the trained logistic regression model.

Chapter 6

Evaluation and Results

This chapter reports how the game was evaluated in a real school environment and how the collected gameplay sessions were later transformed into a dataset for the ADHD screening model. It combines (1) the practical playtesting procedure and participant characteristics, (2) the main quantitative outcomes already presented in the model chapter (recalled here for completeness), and (3) a critical discussion of strengths and limitations.

6.1 Playtesting and Participant Details

Playtesting was conducted at Rambam Elementary School after coordination with the educational staff and approval from the school management. Following internal approval, a consent message was distributed to parents. In order to reduce stigma and prevent expectancy effects, pupils were not informed whether they were selected due to an ADHD-related classification. Instead, students received same-colored stickers in groups of five; within each classroom, one or two sticker colors corresponded to students from the ADHD group without this being disclosed to the children.

This single-blind design is critical for ensuring ecological validity, as it minimizes the ‘Expectancy Effect’ (or Pygmalion Effect), where participants unknowingly alter their performance based on diagnostic labels [14].

Data collection was organized across four age groups (3rd, 4th, 5th, and 6th grades). Each day was dedicated to a single group (either ADHD or non-ADHD), and in each timeslot five students from a class played the game. Across the study period, 104 children participated, approximately one third of whom belonged to the ADHD group. Participation was balanced across grades (approximately 25 students per age group). Gender distribution varied by grade: the 5th–6th grade groups included only girls, while the 3rd–4th grades were mixed (approximately 50/50).

Although the game included vocal instructions, additional guidance from the researcher was required. The written prompts were in English, which many pupils could not read, so before each minigame the researcher explained the goal and the controls again in child-appropriate terms. This observation is important for interpreting behavioral measurements: recorded performance reflects not only the game mechanics but also real-world usability constraints in a school setting.

Finally, the collection protocol was aligned with privacy-by-design principles. The system is designed to upload a minimal session summary to Supabase, while sending only numeric features to the prediction API, avoiding upload of raw personal data or rich media by default.



Figure 6.1: A photo of the classroom setup

6.2 Data Analysis and Findings

From play sessions to a training dataset.

After playtesting, each session was exported at the end of gameplay and persisted through the logging/export routine. The export process produces (1) a compact session JSON for storage and (2) a fixed-length numeric feature vector for model inference. In this build, the Supabase row begins with a one-line human-readable “summary line” containing the feature values in a predefined order.

In parallel, the same values are sent as a 23-element vector to the model API. The model deployment pipeline is reproducible: the classifier is trained as a scikit-learn pipeline, serialized into `adhd_model.pkl`, and exposed via an HTTP endpoint (`POST /predict`) for programmatic inference.

Model performance (recall from Chapter 5).

As reported in the previous chapter, the trained logistic regression model achieved an accuracy above 85% on the held-out test set (see Chapter 5 metrics summary). The confusion matrix interpretation is especially important: in the evaluated split, misclassifications followed a conservative pattern—non-ADHD sessions were predicted as ADHD, rather than the reverse. Practically, this is the “safer” error direction for a screening tool: the worst-case outcome is that a child is referred for a formal diagnostic process, rather than an ADHD-positive case being

missed.

In the context of medical screening theory, this trade-off reflects a prioritization of ‘Sensitivity’ over ‘Specificity.’ For a preliminary screening tool, false positives are acceptable collateral costs to ensure that the False Negative rate (missing a diagnosis) remains close to zero [15].

In addition to overall performance, the selected feature set is behaviorally interpretable and grounded in gameplay events. The summary line includes measures of (a) task performance and safety (e.g., collisions, out-of-bounds losses), (b) timing and variability (e.g., reaction-time median and coefficient of variation), and (c) input dynamics (keypress count and inter-key gap statistics), computed consistently across sessions.

These features were chosen specifically because they can reflect impulsivity, sustained attention, and behavioral variability during short play segments, while remaining easy to compute and deploy at scale.

Qualitative feedback from players.

Beyond numeric metrics, the most consistent qualitative signal was engagement: children reported that they enjoyed the game and many expressed interest in playing it at home. This supports the project’s core design goal—collecting signals unobtrusively in a motivating experience—because high engagement reduces dropout risk and increases the feasibility of repeated measurements over time.

6.3 Strengths and Limitations

Strengths.

1. **Real-world evaluation setting:** Testing in an operational school environment increases ecological validity compared with lab-only collection.
2. **Stigma-reducing procedure:** The sticker-based grouping minimized disclosure and expectancy effects, supporting more natural behavior during gameplay.
3. **Robust, privacy-oriented pipeline:** Session export is compact, with the model receiving numeric features only, and Supabase storing a minimal summary by design.
4. **Reproducible deployment:** The model training and serving flow is explicit (train → serialize → deploy → call /predict), enabling consistent updates and auditability.

Limitations.

1. **Participant diversity and sampling:** Data were collected at a single school and reflect its specific demographics and constraints; broader replication is required for generalization.

2. **Supervision effects:** Many pupils required guidance despite vocal instructions, because written prompts were not accessible (English). This can influence reaction times and error rates and should be addressed in future iterations (e.g., full Hebrew UI, more icon-driven prompts).
3. **Label and scheduling structure:** Collecting “ADHD days” versus “non-ADHD days” simplifies logistics but may introduce contextual differences (fatigue, classroom dynamics, researcher pacing) between groups.
4. **Model decision bias:** The confusion matrix pattern indicates a conservative tendency toward ADHD predictions, which is safer than false negatives but can raise false-positive referrals; future work should calibrate thresholds and validate with additional cohorts.

Chapter 7

Discussion and Conclusions

7.1 Summary of Achievements

This project set out to resolve the “Engagement Gap” inherent in traditional ADHD screening. By integrating the clinical rigor of Continuous Performance Tests (CPTs) with the immersive nature of video games, “TimeSkip” successfully demonstrated a new paradigm for pediatric assessment. The pilot study results (Chapter 6) confirm that the system meets its dual objective:

1. **Ecological Validity:** Children engaged with the platform naturally, reporting high enjoyment levels rather than the stress typical of clinical testing.
2. **Diagnostic Accuracy:** With an accuracy rate of >85% and a “Safe Failure” profile (prioritizing sensitivity), the Logistic Regression classifier proved that gameplay data is a valid proxy for cognitive functioning.

7.2 Bridging Theory and Data

A key finding of this research is the strong correlation between the theoretical models outlined in Chapter 2 and the feature importance discovered by our Machine Learning model in Chapter 5. Specifically, the fact that the Coefficient of Variation (CV) of reaction times emerged as a dominant predictor strongly validates Barkley’s Unifying Theory. It confirms that ADHD is not merely a deficit of “speed,” but a deficit of consistency. The system successfully captured the “micro-behaviors”—the subtle fluctuations in motor control—that are invisible to the human eye but statistically significant to the algorithm.

7.3 Versatility and Future Potential: The “Swords of Iron” Initiative

While the core scope of this report focuses on ADHD, the robust architecture of “TimeSkip” proved to be a powerful foundation for broader mental health applications. Following the outbreak of the “Swords of Iron” war (2023-2024), there was a documented rise in pediatric anxiety and Post-Traumatic Stress across Israel. Recognizing the urgency, our team identified the platform’s potential contribution and expanded its capabilities during an emergency Hackathon.

From Software to Sensor Fusion:

To adapt the system, we expanded the development team to include a mechanical engineer and an additional developer. This allowed us to evolve the project from a purely software-based solution into a multi-modal diagnostic system:

- **Physiological Feasibility:** A standalone hardware prototype was built to demonstrate the concept of sampling Heart Rate Variability (HRV) in real-time. This hardware operated as an external proof-of-concept to validate the signal quality, distinct from the game’s current software loop.
- **Facial Analysis Exploration:** We initiated strategic discussions with industry experts (specifically, the leadership of a Facial Affect Analysis provider) to assess the viability of integrating micro-expression detection. These consultations confirmed the theoretical compatibility of such technologies with our architecture for future iterations.

These explorations successfully demonstrated that “TimeSkip” is not just a game, but a versatile infrastructure capable of evolving into a multi-modal diagnostic hub.

The Holistic Approach:

By fusing these external physiological signals with the game’s native behavioral logs (inhibition latency, motor agitation, and decision-making patterns), we created a comprehensive “diagnostic envelope.” This initiative demonstrated that the original “TimeSkip” engine is not just a standalone game, but a versatile infrastructure. Its solid code base allowed for the seamless integration of external hardware and advanced AI layers, proving that the system can serve as a modular hub for various digital health screening tools beyond its original purpose.

7.4 Implications for Tele-Health

The successful deployment of the system using WebGL and Supabase (Chapter 3) highlights its potential for remote monitoring. Unlike MOXO or TOVA tests that require a clinic visit, “TimeSkip” can be deployed to a child’s home computer. This capability could allow clinicians to monitor a child’s response to medication (e.g., Ritalin) over time, collecting daily data points without logistical burden.

7.5 Conclusion

“TimeSkip” represents a step forward in the democratization of cognitive health. By translating the “black box” of neuropsychological testing into the accessible language of play, the project offers a tool that respects the child’s emotional needs while satisfying the clinician’s need for objective data. While further validation with larger, diverse cohorts is necessary, the current

results suggest that the future of screening lies in systems that are invisible, engaging, and data-driven.

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Project Resources

- Game Itch.io: <https://eran-david.itch.io/timeskip>
- GitHub Repository: <https://github.com/gamedev-ariel/timeskip>
- Scripts Folder: <https://github.com/gamedev-ariel/timeskip/tree/itamar/Assets/Scripts>
- Classifier API: <https://huggingface.co/spaces/ithamarspitz/adhd-noise-classifier>
- Supabase Dashboard: <https://supabase.com/dashboard/org/qnksxfjmlztqhnlaauvz>
- Demo Video: <https://www.youtube.com/watch?v=pa8Azw4GSfM>

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